Progressive Workload Periodization Maximizes Effects of Nordic Hamstring Exercise on Muscle Injury Risk Factors

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Abstract

Severo-Silveira, L, Dornelles, MP, Lima-e-Silva, FX, Marchiori, CL, Medeiros, TM, Pappas, E, and Baroni, BM. Progressive workload periodization maximizes effects of Nordic hamstring exercise on muscle injury risk factors. *J Strength Cond Res* XX (X): 000–000, 2018—This study aimed to investigate the effect of 2 different Nordic hamstring exercise (NHE) training programs (constant workload vs. progressive workload) on multiple risk factors for hamstring strain injury in high competitive level athletes. Twenty-one premier league rugby union players (~26 years) were randomized into 1 of 2 experimental groups and completed an 8-week NHE training program: constant training group (CTG) or progressive training group (PTG). Knee flexor/extensor strength and biceps femoris long head (BF_{LH}) muscle architecture were assessed through isokinetic dynamometry and ultrasonography, respectively, before and after the training programs. Increased hamstring concentric ($\Delta = 8\%$, $p = 0.004$, $d = 0.42$) and eccentric ($\Delta = 7\%$, $p = 0.041$, $d = 0.39$) peak torques, as well as H:Q conventional ($\Delta = 7\%$, $p = 0.015$, $d = 0.54$) and functional ($\Delta = 6\%$, $p = 0.026$, $d = 0.62$) ratios were observed in the follow-up compared with baseline for the PTG group only. Nordic hamstring exercise training significantly increased BF_{LH} muscle thickness (CTG: $\Delta = 7\%$, $p = 0.001$, $d = 0.60$; PTG: $\Delta = 7\%$, $p < 0.001$, $d = 0.54$) and fascicle length (CTG: $\Delta = 8\%$, $p < 0.001$, $d = 0.29$; PTG: $\Delta = 10\%$, $p < 0.001$, $d = 0.35$) in both groups. Training adaptations on hamstring peak torque (concentric and eccentric) and H:Q ratio (conventional and functional) "likely favor" PTG, whereas changes in BF_{LH} fascicle length "possibly favor" PTG. In conclusion, only PTG enhanced hamstring strength; both NHE training programs increased BF_{LH} muscle thickness and fascicle length, but progressive training generated greater changes on fascicle length than constant training. Medical/coaching staff should be aware that NHE with adequate training periodization with workload progression to maximize its benefits.

Key Words eccentric training, hamstring injury, injury prevention, rugby

Introduction

Hamstring strain injury (HSI) is one of the most frequent injuries in team sports such as football (soccer), American football, Australian rules football, rugby union, and rugby league (32). Each injury typically results in around 17 days absence from training and competition, but it is not uncommon for some players to take more than a year to return to play (16). Hamstring strain injury presents high recurrence rates (~16–32%) (32), with the rehabilitation process being more complex and achieving positive outcomes more challenging for reinjured players (19). In addition to the consequences for players' health and career, HSIs can negatively affect team performance (18) and cause financial loss to the club (21). Despite the extensive research on HSI in recent years, HSI incidence has increased by 4% annually in male professional football (16).

Hamstring strain injuries commonly occur during high-speed running, particularly at the terminal swing phase when the hamstrings contract eccentrically to decelerate the extending knee and flexing hip (19). Older age and previous hamstring injury are well accepted nonmodifiable risk factors for HIS (32). At the same time, prospective studies have found that players with a range of intrinsic characteristics such as eccentric hamstring weakness (9,27,33,38), muscle strength imbalance (hamstring-to-quadriceps [H:Q] ratio) (14,27), and reduced biceps femoris long head (BF_{LH}) fascicle length (38) are more prone to sustaining an HSI.

Elite sporting clubs have invested in hamstring conditioning programs using the Nordic hamstring exercise (NHE) to prevent HSIs (29). Nordic hamstring exercise added to the

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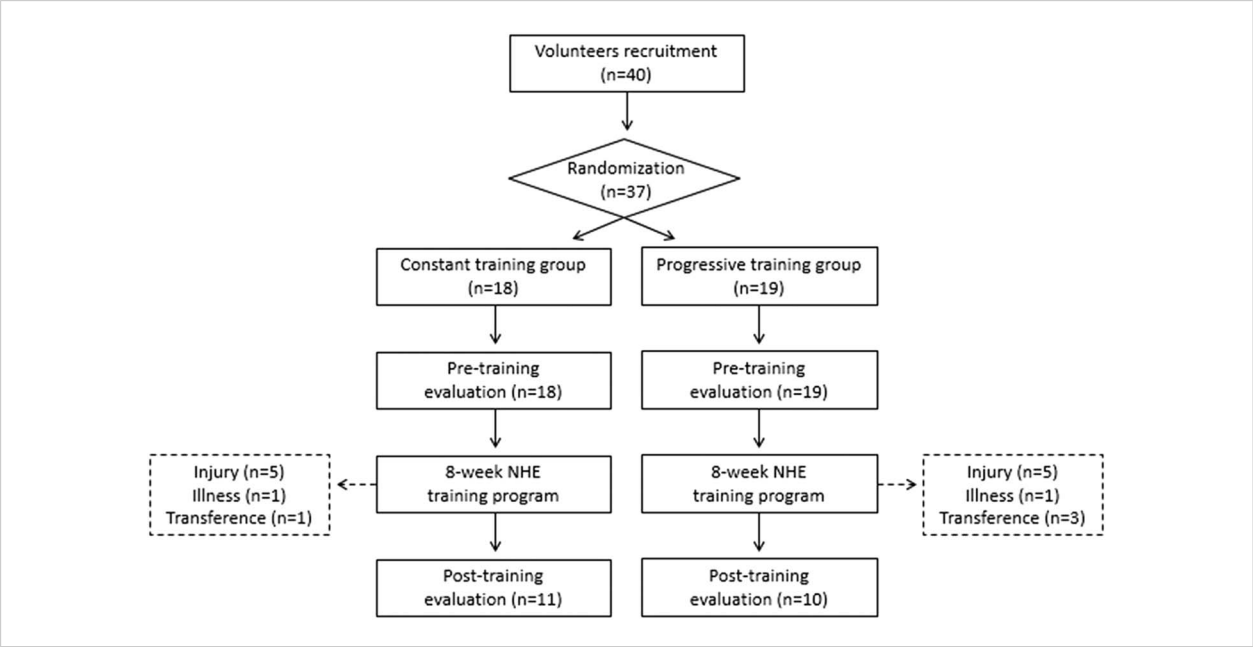

Figure 1. Flowchart of the study. NHE = Nordic hamstring exercise.

training routine of professional athletes reduced the incidence and severity of HSIs (4,11,34,36). This preventive effect has been widely attributed to a significant strengthening of hamstrings (10,23,28,30) and an improved H:Q ratio (10,30), whereas recent studies have also found enhancement of BF_{LH} fascicle length through NHE training programs (2,10,28).

As expected for any kind of resistance training, it seems plausible that muscular adaptations to NHE training (and, consequently, its preventive effect) may be affected by periodization used during the training program (26). However, although some studies did not describe the NHE training periodization (11,36), others have used 3–4 sets of 6–12 repetitions with different workload progressions (2,4,10,23,28,30,34). The injury prevention program developed by the Fédération Internationale de Football Association (FIFA) Medical and Research Center (FMARC), named "FIFA 11+," suggests one set of 3–5 repetitions for beginners, 7–10 repetitions for intermediate, and 12–15 for advanced players (37), but further details about the workload progression are missing. As far as we know, a single study compared the muscular adaptations with different NHE training programs and found that both low- and high-volume NHE training stimulate increases in BF_{LH} fascicle length and hamstring eccentric strength (35). Therefore, the optimal periodization for NHE to produce beneficial changes for the hamstrings and subsequently reduce HSI remains unclear.

The "workload progression" is a principle of physical training that describes the gradual increase of stress placed on the body during the training program, and it is considered a key factor for optimizing training responses (26). However, the impact of workload progression on NHE training programs remains unexplored. Therefore, the aim of this study was to investigate the effect of 2 different NHE training programs (constant workload vs. progressive workload) on multiple risk factors for HSI in high competitive level athletes.

Methods

Experimental Approach to the Problem

A randomized trial was designed to determine the effects of an 8-week NHE training program on HSI risk factors. Subjects were randomized into 2 experimental groups: constant training group (CTG) or progressive training group (PTG). A control group (without training) was not included in this study because the club required that all players should be engaged in a preventive program using the NHE. The NHE training sessions were performed in the team's training field, and the evaluations were conducted at the Physiotherapy Laboratory of the Federal University of Health Sciences of Porto Alegre (Brazil).

Subjects

The study proposal was presented and approved by the head manager and the coaching staff of a national premier league rugby union team based in Porto Alegre (Brazil). Players usually performed 3 weekly practices on the field and 2–3 strength and conditioning sessions at the gym or the athletic track. Throughout the 8-week training program, the team played 4 matches for the state championship.

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To be included in the study, players had to be at least 18 years old, with a minimum of 2 years of experience playing competitive rugby, and regularly participating in the team's training routine during the data collection period. The age range was between 19 and 33 years old. The following exclusion criteria were applied: (a) history of muscle injury in the posterior thigh up to 6 months before the beginning of the study; (b) current musculoskeletal injury or any illness that could impair the execution of NHE or isokinetic testing; and (c) noncompliance with the full NHE training program. The study was approved by the Institutional Ethics Committe of Federal University of Health Sciences of Porto Alegre (protocol number: 63261516.8.0000.5345). Subjects provided written and informed consent.

In a 40-player rugby union squad, 37 players met the inclusion/exclusion criteria and agreed to participate through a signed consent form. During the study period, 10 players sustained musculoskeletal injuries during games or practices, but none during the NHE execution. A single HSI was recorded in the CTG. In addition, 2 players got sick and 4 were transferred to other clubs. Therefore, 21 players completed the NHE training program properly and were included in this study: 11 in the CTG and 10 in the PTG (Figure 1). Constant training group and PTG were similar in age (27.2 ± 3.26; 25.2 ± 3.34; $p = 0.903$), body mass (90.1 ± 14.3; 88.6 ± 12.8; $p = 0.666$), and height (1.75 ± 0.05; 1.76 ± 0.08; $p = 0.435$).

Procedures

Randomization and Blinding. The randomization process was conducted through the website www.random.org by an assistant not involved with assessment or training. Researchers involved in NHE training did not attend evaluation sessions and researchers responsible for evaluations did not follow the training sessions.

Outcomes and Follow-Up. Knee flexor/extensor strength and BF_{LH} muscle architecture were assessed through isokinetic dynamometry and ultrasonography, respectively. Evaluations were performed within 1 week before the start and within 1 week after the end of the training programs. Hamstring eccentric peak torque and BF_{LH} fascicle length were the primary outcomes of the study. The secondary outcomes were: hamstring and quadriceps concentric peak torques; H:Q conventional and functional ratios; BF_{LH} muscle thickness; and pennation angle.

Muscle Architecture Assessment. A standardized procedure was used for ultrasound assessment (10). A B-mode ultrasonography system (Vividi; GE Medical Systems, Fairfield, CT) with a linear-array probe (GE 8L; frequency: 13 MHz; depth: 6 cm; width: 4 cm) was used to assess muscle architecture parameters of BF_{LH} by an experienced examiner. After at least 10 resting minutes, subjects were assessed in prone position with the hips at neutral position, the knees fully extended, and the muscles relaxed. Biceps femoris long head length was determined as the distance from the ischial tuberosity to the superior border of the fibular head, and the ultrasound scans were taken at 50% of the muscle length. The ultrasound probe was positioned perpendicularly to the skin in the muscle's longitudinal axis, with slight adjustment in probe orientation as necessary to optimize the aponeuroses and fascicle identification.

Three ultrasound images in each lower limb were taken and stored for analysis through the ImageJ software (National Institutes of Health, USA). Muscle thickness, pennation angle, and fascicle length from each image were assessed and the average value of the 3 images was used for statistical analyses. Muscle thickness was calculated as the mean length of 3 mutually parallel lines perpendicular to the superficial and deep aponeurosis. The most visible fascicle in each image was used for measurement of fascicle pennation angle and estimation of fascicle length using a validated equation (24).

Muscle Strength Assessment. A standardized testing protocol was used for isokinetic assessment (3). After a 5-minute warm-up exercise in a cycle ergometer with self-selected load and cadence, players were properly positioned in the isokinetic dynamometer Biodex System 4 Pro (Biodex

Week	CTG	PTG
1	2 x 6	
2	3 x 6	
3	3 x 6	3 x 8
4	3 x 6	3 x 10
5	3 x 6	4 x 8-10
6	3 x 6	4 x 8-10
7	3 x 6	4 x 10
8	3 x 6	4 x 10

Figure 2. Execution of the Nordic hamstring exercise (NHE) and training periodization (number of sets and repetitions along the 8-week program) for the constant training group (CTG) and the progressive training group (PTG).

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Table 1. Muscle architecture and strength of constant training and progressive training groups at pre- and post-training evaluations (mean ± *SD*).*

	Constant training group–CTG ($n = 10$)						Progressive training group–PTG ($n = 11$)					
	Pre	Post	Δ%	*p*	ES		Pre	Post	Δ%	*p*	ES	
Muscle architecture												
BF_{LH} muscle thickness (cm)	2.27 ± 0.24	2.42 ± 0.27	6.61	0.001†	0.60‡		2.38 ± 0.32	2.56 ± 0.36	7.56	<0.001†	0.54‡	
BF_{LH} pennation angle (°)	15.65 ± 3.36	15.60 ± 3.30	−0.32	0.217	0.02		15.96 ± 3.42	15.92 ± 3.35	−0.25	0.528	0.01	
BF_{LH} fascicle length (cm)	10.48 ± 2.74	11.26 ± 2.83	7.44	<0.001†	0.29§		11.13 ± 2.83	12.26 ± 3.20	10.15	<0.001†	0.35§	
Muscle strength												
Quadriceps CON PT (N·m)	275.50 ± 27.17	276.60 ± 24.91	0.40	0.761	0.04		278.05 ± 48.86	280.85 ± 53.95	1.01	0.619	0.06	
Hamstrings CON PT (N·m)	142.25 ± 19.66	144.74 ± 20.98	1.75	0.250	0.13		146.64 ± 24.31	157.95 ± 30.48	7.71	0.004†	0.42§	
Hamstrings ECC PT (N·m)	204.58 ± 43.32	207.01 ± 41.67	1.19	0.516	0.06		211.17 ± 31.81	225.64 ± 43.29	6.85	0.041†	0.39§	
H:Q conventional ratio	0.52 ± 0.05	0.52 ± 0.05	0.00	0.406	0.00		0.53 ± 0.07	0.57 ± 0.08	7.55	0.015†	0.54‡	
H:Q functional ratio	0.74 ± 0.14	0.75 ± 0.12	1.35	0.671	0.08		0.76 ± 0.06	0.81 ± 0.10	6.58	0.026†	0.62‡	

*Δ% = mean percent change; ES = effect size; BF_{LH} = biceps femoris long head; CON = concentric; PT = peak torque; ECC = eccentric; H:Q = hamstring-to-quadriceps.
†$p < 0.05$.
‡Moderate effect size.
§Small effect size.

Medical Systems, Shirley, NY). Calibration of the equipment and gravitational correction were performed. Subjects received standardized instructions about the testing protocol and performed a specific warm-up (10 submaximal repetitions at $180° \cdot s^{-1}$). Thereafter, volunteers performed 2 sets of 3 consecutive maximum contractions of flexion-extension in concentric-concentric mode and 2 sets in eccentric-eccentric mode at $60° \cdot s^{-1}$. Subjects were verbally encouraged during tests to produce maximal strength with a 1-minute rest between each test.

The highest peak torque value obtained in each evaluation mode was used for statistical analysis. In addition, we calculated the conventional ratio (knee flexor concentric peak torque divided by knee extensors concentric peak torque) and the functional ratio (knee flexor eccentric peak torque divided by knee extensors concentric peak torque) (6).

Nordic Hamstring Exercise Training Program. Before the start of the training program, we provided instructions on the NHE and a demonstration of its correct execution. Players were instructed to use their maximal effort to resist the forward-falling motion using the hamstring muscles. A controlled movement for ~3 seconds was the aim in each repetition. Participants were instructed to use their arms and hands to buffer the fall and immediately get back to the starting position by pushing up with their arms (30).

Training sessions were performed twice a week (Tuesdays and Thursdays) for an 8-week period (total of 16 training sessions). All training sessions were executed after a proper warm-up before field practice. Two researchers were present during every training session to instruct each experimental group. They ensured correct execution of the NHE, and number of sets and repetitions performed by players in CTG and PTG (see periodization in Figure 2). When a player did not attend the practice session, he performed the NHE protocol by himself and this was confirmed be sending a video through an instant messages app to researchers. Therefore, all players complied with the full NHE training program in this study.

Statistical Analyses

Sample size was a priori calculated using the primary outcomes (i.e., hamstring eccentric peak torque and BF_{LH} fascicle length) of a similar study (10). Power was set at 80% with an alpha level of 0.05, returning a sample size of 10 subjects per group.

Independent *t*-test was used to compare groups for age, body mass, and height. Data from both legs were combined (20 legs for CTG and 22 legs for PTG) to assess training effects on muscle architecture and strength outcomes.

Within-group analysis to verify training effects were performed with the following calculations: mean percent

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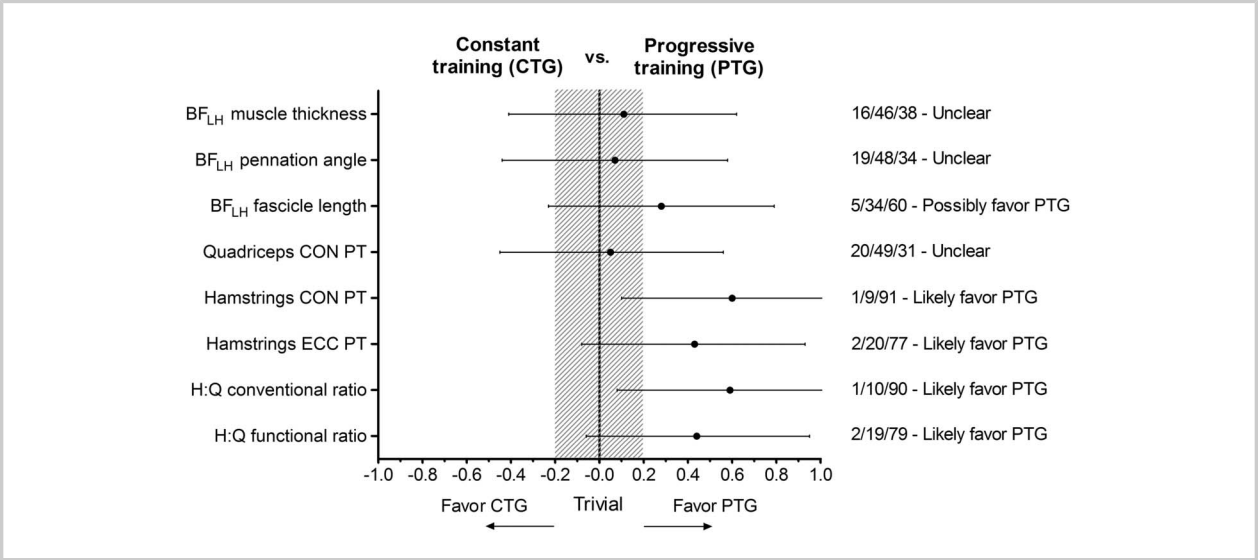

Figure 3. Between-group analysis: constant training group (CTG) and progressive training group (PTG). The forest plot presents the effect size and 95% CI. Left column presents the quantitative chances of substantial effects for CTG, trivial effects, and substantial effects for PTG, as well as the magnitude-based inference (see methods for details). BF_{LH} = biceps femoris long head; CI = confidence interval; CON = concentric; PT = peak torque; ECC = eccentric; H:Q = hamstring-to-quadriceps.

change (Δ% = post-training/pre-training − 1); paired-sample *t*-test, with a significance level set as 5% ($\alpha < 0.05$); effect size (ES) through the Cohen's *d* ($ES = [M_{post} - M_{pre}]/SD_{pooled}$, where M_{post} is the mean post-training measure, M_{pre} is the mean pre-training measure, and SD_{pooled} is the pooled *SD* of the pre- and post-measurement), and training effects were considered as "trivial" (ES < 0.2), "small" (ES > 0.2), "moderate" (ES > 0.5), or "large" (ES > 0.8) (13).

Longitudinal percent changes (pre- to post-training) were used for between-group analysis. Data were compared through independent-sample *t*-test and analyzed for practical significance using magnitude-based inferences because traditional statistical approaches often do not indicate the magnitude of an effect, which is typically more relevant to sports medicine than statistical significance (22). The chances of a possible substantial effect favorable to CTG or PTG were calculated (i.e., greater that the smallest worthwhile change [0.2 multiplied by the between-subject *SD*]). Quantitative chances of trivial effects and substantial effects for each group were assessed qualitatively as follows: <1%, almost certainly not; 1–5%, very unlikely; 5–25%, unlikely; 25–75% = possibly; 75–95% = likely; 95–99% = very likely; and >99% = almost certain (8). When the CTG and PTG values were both >5%, the inference was classified as unclear (8).

Results

Table 1 presents the within-group analysis. Nordic hamstring exercise training significantly increased BF_{LH} muscle thickness (moderate ESs) and fascicle length (small ESs) in both groups, whereas the BF_{LH} pennation angle did not change in either group (trivial ESs). Increased hamstring concentric and eccentric peak torques (small ESs) as well as H:Q conventional and functional ratios (moderate ESs) were observed in the follow-up compared with baseline for the PTG group only. None of the strength measures changed within the CTG (trivial ESs).

Between-group comparison through null hypothesis tests demonstrated no significance difference between groups for any outcome ($p > 0.05$). Practical significance analysis is presented in Figure 3. Magnitude-based inference supports that CTG and PTG had similar adaptations in muscle thickness, pennation angle, and quadriceps concentric peak torque. However, training adaptations on BF_{LH} fascicle length "possibly favor" PTG (60% chance), whereas concentric and eccentric peak torques "likely favor" PTG (91 and 77% chance, respectively) as well as H:Q conventional and functional ratios (90 and 79% chance, respectively).

Discussion

This is the first study to directly compare the effects of constant and progressive training programs using the NHE on risk factors for HSI. The main findings were that: (a) only the PTG resulted in significant knee flexor strength increases and improved H:Q ratios; and (b) the PTG had more pronounced changes on BF_{LH} fascicle length, although both training programs increased BF_{LH} thickness. Our results highlight the importance of following the principle of workload progression in NHE training programs to optimize changes in the knee flexors that may be protective of HSI.

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More pronounced differences between the 2 groups were found for isokinetic strength and H:Q ratios where only the PTG had significant changes. Studies on physically active volunteers (10) and professional football players (23,30) have shown 11–13% improvement in eccentric strength through NHE training programs of 4–13 weeks. These improvements were slightly higher than those observed in the current study (~7%), which may be related to the conditioning level and the baseline hamstring strength of rugby players engaged in our study. Poor eccentric strength has been suggested as an HSI risk factor for decades (20), and prospective studies with football (27,38), Australian football (33), and rugby union (9) players found that low levels of eccentric strength at pre-season predispose athletes to HSI during the season. The current study demonstrated that it is possible to increase hamstring strength in high competitive level athletes by adding a feasible NHE program to the training routine of a premier league rugby union team.

The role of H:Q ratio as a risk factor for HSI has been debated over the years. In contrast to a prospective study that reported no association between H:Q strength ratio and HSI (15), others support that athletes with agonist-antagonist strength imbalances are more prone to sustain a muscle injury (14,27). The 0.6 cutoff for the H:Q conventional ratio was proposed in the 1960s by Klein and Allman (25), and this reference value gained strength after Heiser et al. (20) demonstrated a reduction in the number of injuries in American football players through a training program that used this cutoff. From the late 1980s, researchers became increasingly interested in H:Q ratio that comprised eccentric knee flexor and concentric knee extensor muscle actions, and the 1.00 cutoff has been commonly used to assess the functional strength imbalance in athletes (1). In this study, although quadriceps strength did not change, both concentric and eccentric hamstring strength improved in the PTG, resulting in improvements in both the H:Q conventional and functional ratios. Although rugby players did not reach the reference values suggested in the literature (6), their H:Q ratios moved in direction to more appropriate levels of strength muscle balance with progressive training periodization.

Biceps femoris long head fascicle length has been recently added to the list of potential risk factors for HSI. A case-control investigation reported that athletes with a unilateral HSI history have shorter BF_{LH} fascicles in the previously injured limb than in the contralateral uninjured limb (40), whereas a prospective study demonstrated that professional football players with shorter BF_{LH} fascicles (<10.56 cm) were 4.1 times more likely to sustain a future HSI than those with longer fascicles (38). These findings provide strong arguments for using BF_{LH} ultrasonography for screening tests for muscle injuries in sports. The mechanism by which shorter fascicles predispose a player to injury is not fully understood, but a plausible hypothesis is that shorter fascicles (consequently, fewer in-series sarcomeres) may be more susceptible to overstretch and subsequent damage during intense eccentric contractions (31), such as those performed during the terminal swing phase of high-speed running (19).

Eccentric training induces addition of serially arranged sarcomeres (sarcomerogenesis) in animal models (12), whereas human studies have reported significant increases in fascicle length after eccentric training programs for hamstrings (39) and other muscle groups (7). Recent trials have demonstrated that NHE training is capable of increasing BF_{LH} fascicle length by 13–24% in recreationally active populations (2,10,28). The competitive rugby players engaged in our study had fascicle length mean increases of ~7 and ~10% in CTG and PTG, respectively. The conditioning status of rugby players perhaps explains the smaller percent changes compared with nonathletes from previous studies. In absolute terms, changes observed in our study represented a ~0.78-cm longer fascicle in CTG and a ~1.13-cm longer fascicle in PTG. Considering that the probability of HSI is reduced by ~21% for every 1-cm increase in fascicle length (38), these changes may theoretically help to prevent HSI in rugby players. In addition, although both NHE training programs resulted in BF_{LH} fascicle increases, our findings support that better results can be expected when players follow progressive training periodization.

To the best of our knowledge, the study by Presland et al. (35) was the only previous study to explore the effect of different training programs on muscle adaptation to NHE. That study reported a similar increase in hamstring eccentric strength and BF_{LH} fascicle length after 6 weeks of low-volume (8 repetitions per week) or high-volume (up to 100 repetitions per week) NHE training (35). However, the Presland study assessed the effect of NHE training volume in recreationally active men, whereas the current study investigated the effect of a moderate-volume NHE training progression in well-trained rugby players; then, comparison between studies should be made with caution. Interestingly, Presland's findings suggest that training volume is not a key factor for muscle adaptation to NHE training, and our results support the importance of training progression (and, consequently, training volume) on hamstring adaptation to an NHE program. Further research is needed to elucidate that issue.

The growing body of evidence demonstrating its effectiveness has been accompanied by adoption of NHE by elite team sports (29). However, the periodization of NHE training used by competitive teams along the season may be influenced by several factors. Congestion periods, travel, coaching staff compliance, and even the team's performance on competition sometimes impair an optimal periodization schedule. Bahr et al. (5) showed that the majority of premier league football teams are not fully compliant to an evidence-based hamstring injury prevention program of 3 weekly sessions for 10 weeks, followed by a once a week maintenance period along the season (4,34). A systematic review demonstrated that the efficacy of the NHE and other eccentric

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exercises for prevention of HSIs depend on intervention compliance (17). Therefore, the medical/coaching staff should be concerned not only with using the NHE but also with using adequate training periodization to maximize its benefits.

A progressive increase of exercise intensity is also considered a key factor to optimize the muscular responses to resistance training (26). The addition of external load, such as holding a plate, may be used to increase the intensity of NHE. However, this is not always a feasible approach when the NHE is performed on the field. Moreover, it is important to note that as a player improves his movement execution throughout the training program, he can lower his torso in a controlled manner by a greater range of motion (i.e., the point of failure is getting closer to the ground). Consequently, the lever arm also increases progressively along the training program, thus enhancing hamstring strength required during the movement execution. Therefore, we recommend that practitioners stay alert to how players perform the NHE. If a player does not perform the exercise maximally and simply lets the torso fall at the beginning of the discomfort, the progression of exercise intensity will be compromised.

The high number of players excluded due to injury reduced considerably the sample size in this study. However, this is expected when researchers develop a randomized trial into the real world of competitive rugby, where teams frequently lose several players due to injury throughout the season. Interestingly, only one player (allocated to the CTG) sustained an HSI during the study period (~10 weeks). The main limitation of the current study was that HSI incidence could not be followed along the entire season. This continuous monitoring would allow us to verify whether the specific muscular adaptations of the progressive training reduced HSI incidence. Although the difficulties in following player exposure and injuries along the entire season are acknowledged, future studies comparing NHE protocols and encompassing this long-time approach should be encouraged.

In conclusion, only PTG enhanced concentric and eccentric hamstring strength, consequently improving both H:Q conventional and functional ratios. Both NHE training programs used in the current study increased BF_{LH} muscle thickness and fascicle length, but PTG generated greater changes on fascicle length than constant training. These findings support the importance of following the workload progression in NHE programs to maximize the training benefits in high competitive level athletes.

Practical Applications

This study demonstrated that following a workload progression in NHE programs is important to maximize the training benefits in high competitive level athletes. Medical/coaching staff should be concerned not only with using the NHE but also with using adequate training periodization with workload progression to maximize its benefits. Sport scientists should investigate the influence of other periodization parameters such as weekly frequency and volume per session (number of sets and repetitions per set) on the muscular adaptation to NHE training.

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